Sheroz Shahid Tabassum

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github.com/sherozshahid11 | scholar.google.com/citations | orcid.org/0009-0007-8988-6969



ABOUT ME

Researcher specializing in Battery Management System (BMS) and Lithium-ion technologies, focusing on utilizing AI techniques for accurate State Estimation. Expert in Advanced Data-Driven Solutions to enhance performance and reliability. My work leverages Responsible AI to diagnose model behavior and optimize algorithmic efficiency, balancing the trade-off between computational simplicity and predictive accuracy. With a strong background in mathematical modelling, I aim to refine the intersection of machine learning and multi-objective optimization for real-time applications.

PROFESSIONAL EXPERIENCE

TEACHER ASSISTANT, *Industrial AI and IoT (IAI) Research Group* — 09/2025 – Present, Islamabad , Pakistan
- As a Teaching Assistant, I guide undergraduate students in Pakistan through AI and IoT projects, deploying in physics-constrained settings like energy storage

INTERN ENGINEER, *T.M Rubber (Pvt) Ltd, Lahore Pakistan* — 05/2022 – 09/2022, Lahore, Pakistan
- Worked on an Automated Rubber Tapping System to improve efficiency and reliable operation in the rubber processing workflow.
- Collaborated with the engineering team on automation and control tasks, gaining hands-on experience in industrial system development.

UNDERGRADUATE RESEARCH STUDENT, *Programming Fundamentals Lab, COMSATS Islamabad* — 10/2020 – 06/2021, Sahiwal, Pakistan
- Developed a Library Management System at the Programming Fundamentals Lab in C++ to manage book records, student data, and borrowing operations efficiently.

EDUCATION

M.E IN MECHATRONICS ENGINEERING, *National University of Sciences and Technology (NUST)* — 09/2024 – 05/2026, Islamabad, Pakistan
Thesis: *Explainable AI-guided optimization of data-driven battery state estimation*
- Intelligent Control in Electrochemical System
- Machine Learning
- Paradigms of Artificial Intelligence
- Advance Embedded Systems & Data Acquisition

B.E. IN MECHANICAL ENGINEERING, *COMSATS University* — 09/2019 – 07/2023, Sahiwal, Pakistan
Thesis: *Design and Fabrication of an Automatic Tyre Inflation System for Parking Facility*
- Mechanics of Materials & Solid Mechanics
- Machine Design & Drafting (CAD)

CERTIFICATES

Supervised Machine Learning: Regression and Classification Stanford Online | MACHINE DESIGN PART I |
Engineering Virtual Experience Program

PROJECTS

Explainable AI-guided optimization of data-driven battery state estimation, *M.E Thesis*
- Developed a Physics-Informed Numeric Transformer (PINT) for efficient and accurate processing of battery data.
- A 2RC-ECM observer is utilized to decouple multiple physical features from the measured data. Features will then be provided to the advanced numeric transformer to enhance its learning of the battery's internal dynamics, thereby improving performance
- Leveraged Explainable AI (SHAP) as a diagnostic tool to quantify feature sensitivity.
- Formulating the estimation task as a multi-objective optimization problem to resolve the trade-off between algorithmic simplicity and predictive reliability.

Design and Fabrication of an Automatic Tyre Inflation System for Parking Facility, *B.E Thesis*
- The project is a startup by COMSATS
- Design and Fabrication of an Automatic tyre Inflation system include a watch on and modifying the pressure levels as necessary, we examine its part in ensuring proper tyre pressure. The system does this to avoid under- and over-inflation.
- Used SolidWorks for a compatible design, performed tests to verify the effectiveness of the designed Inflation system.

PUBLICATIONS

Generative Explainable AI for Reliable State Estimation of Battery System, 2026
Hackathon IEEE International Conference on Responsible Artificial Intelligence ,IECON
- Two interacting agents integrated with XAI to estimate battery state of charge and internal core temperature.
- Then we introduce a "Reliability Agent" to calculate a "Reliability Factor" continuously in real time to detect drift, generate a diagnostic report, and keep the battery safe. (Accepted; the manuscript will be added soon)

Authors: Sheroz Shahid Tabassum, Hafiz Muhammad Jawaad, M.saeed, Arash Khalatbarsoltani, Umar Shahbaz Khan

XAI-Enhanced Physics-Informed Neural Networks for State Estimation of Li-ion Batteries, 03/12/2025
5th International Conference on Digital Futures and Transformative Technologies (ICoDT2)
Sheroz Shahid Tabassum, Hafiz Muhammad Jawaad, M. Saeed, Hassan Elahi, Arash Khalatbarsoltani, Ayesha Zeb ,Umar Shahbaz Khan, "XAI-Enhanced Physics-Informed Neural Networks for State Estimation of Li-ion Batteries," in 2025 5th International Conference on Digital Futures and Transformative Technologies (ICoDT2).DOI:10.1109/ ICoDT269104.2025.11360728

Model Bias-aware Battery Fast Charging, 07/11/2025
5th International Conference on Digital Futures and Transformative Technologies (ICoDT2)
Hafiz Muhammad Jawaad, **Sheroz Shahid Tabassum**, M. Saeed, Hassan Elah, Arash Khalatbarsoltani, Hamid Jabbar and Umar Shahbaz Khan, "Model Bias-aware Battery Fast Charging," in 2025 5th International Conference on Digital Futures and Transformative Technologies (ICoDT2).DOI:10.1109/ICoDT269104.2025.11360701

A Critical Review of Battery State Estimation: from Black Box to Physics Guided Explainable AI, 2026
Engineering Applications of Artificial Intelligence
Sheroz Shahid Tabassum, M. Saeed, Hafiz Muhammad Jawaad, , Arash Khalatbarsoltani, Ayesha Zeb, Hassan Elahi, Umar Shahbaz Khan, Qadeer Ahmad, and YIJUN He "A Critical Review of Battery State Estimation: from Black Box to Physics Guided Explainable AI," Engineering Applications of Artificial Intelligence, (under-review).

SKILLS

- **AI/ML:** Physics-Informed Neural Networks (PINNs), Explainable AI (XAI), Supervised Learning (Regression/Classification). SHAP (Feature Quantification & Optimization), Time-Series Forecasting, Model Bias-Aware Learning, Manifold Learning (Riemannian Covariance)
- **Programming & Computational Tools:** C/C++ and Python Programming (Expert in developing intelligent algorithms), Advanced Embedded Systems, Applied & mathematical modelling of dynamical systems, SolidWorks (3D Modelling), Battery Characterization (Expert in parametrization and testing of LIBs using HPPC and EIS), COMSOL Multiphysics, MATLAB/SIMLINK (Control Strategy), Prompt Engineering

NETWORKS AND MEMBERSHIPS

Umar Shahbaz Khan, *Ph.D., Professor, National University of Sciences and Technology Islamabad]*

Hassan Elahi, *Ph.D., Associate Professor National University of Sciences and Technology Islamabad*

Hamid Jabbar, *Ph.D., Professor, Head of Department, National University of Sciences and Technology Islamabad.*

Ayesha Zeb, *Ph.D., Assistant Professor, National University of Sciences and Technology Islamabad*

Muhammad Saeed, *Ph.D. battery modeling Research Fellow with Shanghai Jiao Tong University.*

Member IEEE

LANGUAGES

English — Fluent
C1 level

URDU — Native/Bilingual
Mother language

AWARDS

CUI Tech Fest – COMSATS University Islamabad
Winner at Tech Fest, involving a problem solving skills in C++

Best Project Award, organized by ASME
Got 1 Prize by ASME for the Fabrication of Automatic Tyre Inflation For Parking Facility in University

Innovative & Collaborative Research Grant (ICRG)
Secured HEC Grant for my Final Thesis " Manufacturing and Fabrication of Automatic Tyre Inflation System"

OPEN Mic Competition – National University of Sciences and Technology
Got 1 Position, Awarded with Best Singer